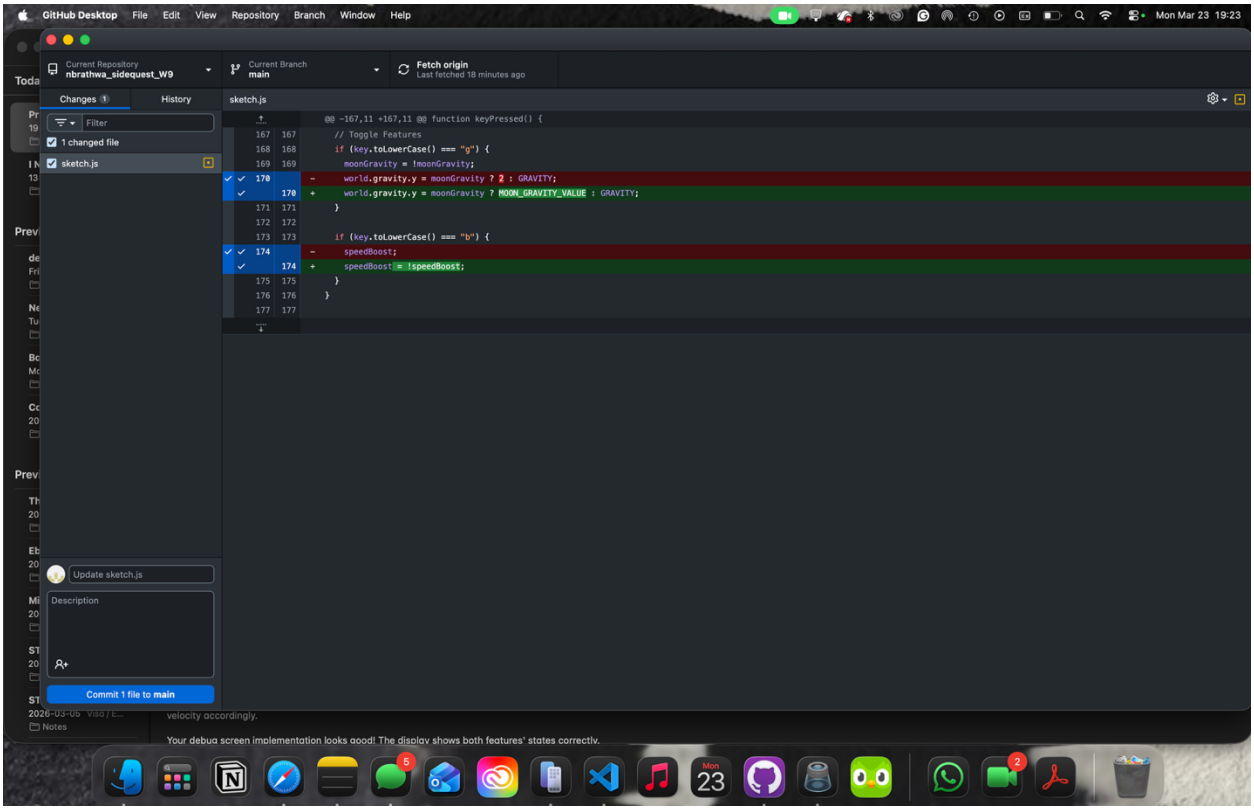
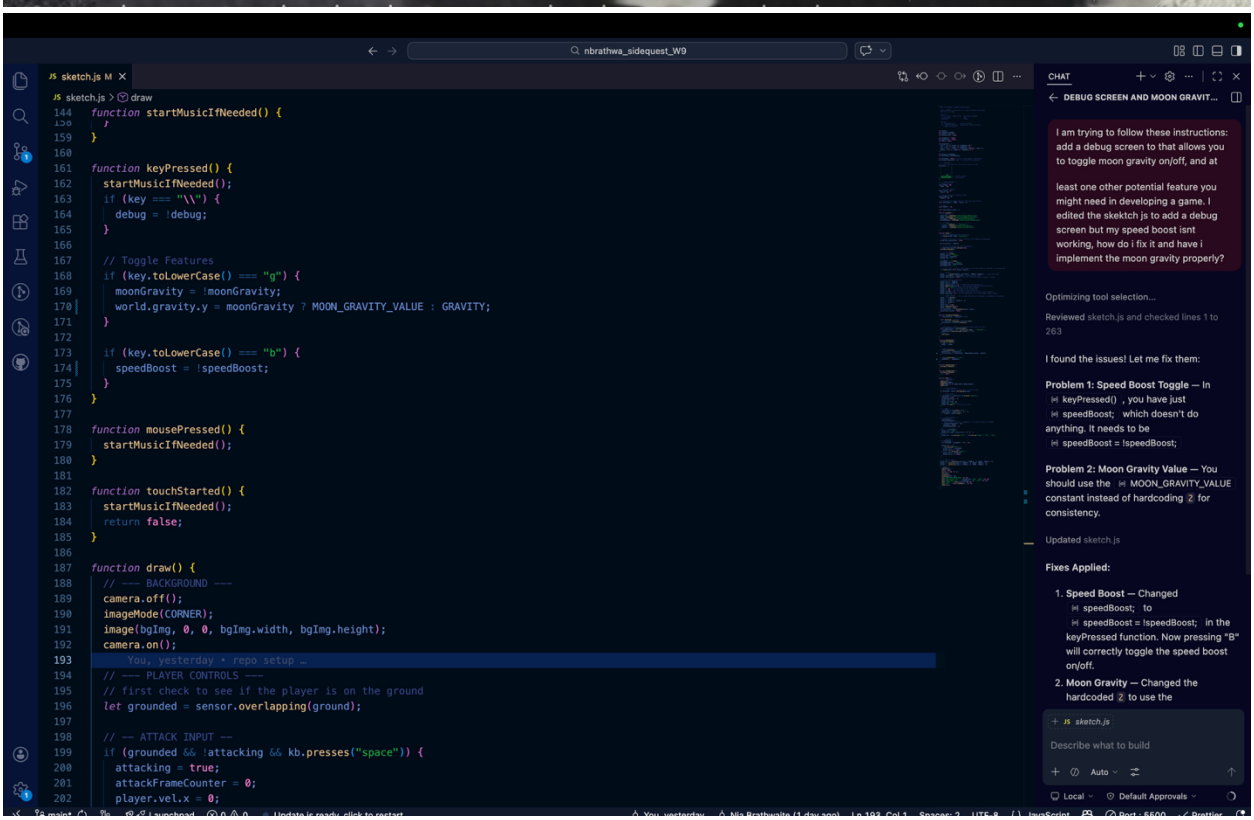
Process & Decision Documentation

Side Quests and A4 (Individual Work)

I added a Moon Gravity Toggle, and I decided that my secondary feature would be a speed boost, so I implemented that. I chose a speed boost because it's common in most games, and I figured it would be simple to implement. It made the player (fox) move twice as fast across the screen.

Role-Based Process Evidence





GitHub Desktop File Edit View Repository Branch Window Help Sun Mar 22 09:49

Current Repository nbrathwa_sidequest_W9 | Current Branch main | Fetch origin Last fetched just now

Changes 1 | History

Filter

1 changed file

sketch.js

sketch.js

```diff
@@ -232,10 +232,10 @@ function draw() {
 
     player.vel.x = 0;
     if (kb.pressing("left")) {
-      player.vel.x = -1.5;
+      player.vel.x = -moveSpeed;
       player.mirror.x = true;
     } else if (kb.pressing("right")) {
-      player.vel.x = 1.5;
+      player.vel.x = moveSpeed;
       player.mirror.x = false;
     }
   }
```

Update sketch.js

Description

Commit 1 file to main

GitHub Desktop File Edit View Repository Branch Window Help Sun Mar 22 09:48

Current Repository nbrathwa_sidequest_W9 | Current Branch main | Fetch origin Last fetched 29 minutes ago

Changes 1 | History

Filter

1 changed file

sketch.js

sketch.js

```diff
   }

   if (key.toLowerCase() === "b") {

@@ -242,32 +242,21 @@ function draw() {

   // --- KEEP IN VIEW ---
   player.pos.x = constrain(player.pos.x, FRAME_W / 2, VIEWW - FRAME_W / 2);
+  player.y = constrain(player.y, FRAME_H / 2, VIEWH - FRAME_H / 2);
+  player.x = constrain(player.x, FRAME_W / 2, VIEWW - FRAME_W / 2);

   if (debug) {
     camera.off();
-
-    fill(0);
-    noStroke();
+    fill(0, 180);
     rect(5, 5, 140, 75, 4);
-
-    // Draw Text
     fill(255);
     textSize(8);
     textAlign(LEFT, TOP);
     text("DEVELOPER DEBUG MENU", 10, 10);
-    text("----------------------", 10, 18);
-
-    // Status indicators
-    fill(moonGravity ? "#228B22" : "#C41E3A");
     text("Moon Gravity (G): " + (moonGravity ? "ON" : "OFF"), 10, 30);
-
-    fill(speedBoost ? "#228B22" : "#C41E3A");
     text("Speed Boost (B): " + (speedBoost ? "ON" : "OFF"), 10, 42);
-
-    fill(255);
-    text("Player X: " + floor(player.x), 10, 54);
+    text("Y-Pos: " + floor(player.y), 10, 54);
     text("FPS: " + floor(frameRate()), 10, 66);
-
     camera.on();
   }
 }
```

Update sketch.js

Description

Commit 1 file to main

GitHub Desktop File Edit View Repository Branch Window Help Sun Mar 22 09:18

Current Repository nbrathwa_sidequest_W9 | Current Branch main | Fetch origin Last fetched 23 minutes ago

Changes 1 | History

Filter

1 changed file

sketch.js

```diff
@@ -20,6 +20,10 @@ let playerImg, bgImg;
 let jumpSfx, musicSfx;
 let musicStarted = false;

+ let speedBoost = false;
+ let moonGravity = false;
+ let debug = false;
+
 let playerAnis = {
   idle: { row: 0, frames: 4, frameDelay: 10 },
   run: { row: 1, frames: 4, frameDelay: 3 },
@@ -65,6 +69,8 @@ const MAP_START_Y = VIEWH - TILE_H * 4;
 // gravity
 const GRAVITY = 10;

+ const MOON_GRAVITY_VALUE = 2;
+
 function preload() {
   // --- IMAGES ---
   playerImg = loadImage("assets/foxSpriteSheet.png");
@@ -154,6 +160,19 @@ function startMusicIfNeeded() {

 function keyPressed() {
   startMusicIfNeeded();
+  if (key === "\\") {
+    debug = !debug;
+  }
+
+  // Toggle Features
+  if (key.toLowerCase() === "g") {
+    moonGravity = !moonGravity;
+    world.gravity.y = moonGravity ? MOON_GRAVITY_VALUE : GRAVITY;
+  }
+
+  if (key.toLowerCase() === "b") {
+    speedBoost;
+  }
 }
```

Update sketch.js

Description

Commit 1 file to main

Safari File Edit View History Bookmarks Develop Window Help Mon Mar 23 20:38

p5js.org

(3) The Biggest Heist in US History: Why Jim... | Google Calendar - March 2026 | Online Shopping Canada: Everyday Low Pric... | Meet - yej-vdtg-joh | speed | Show tab overview

p5.js

Reference
Tutorials
Examples
Contribute
Community
About

English | Accessibility | Search

Reference > speed()

speed()

Sets the audio/video playback speed.

The parameter, `val`, is optional. It's a number that sets the playback speed. 1 plays the media at normal speed, 0.5 plays it at half speed, 2 plays it at double speed, and so on. -1 plays the media at normal speed in reverse.

Calling `media.speed()` returns the current speed as a number.

Note: Not all browsers support backward playback. Even if they do, playback might not be smooth.

Start Coding

Donate

Reference

DOM
addClass()
attribute()
center()
child()
class()
doubleClicked()
dragLeave()
dragOver()
draggable()
drop()
elt
hasClass()
height
hide()
html()
id()
mouseClicked()
mouseMoved()
mouseOut()
mouseOver()

Examples

Speed: 0.5

```
let dragon;

function setup() {
  createCanvas(100, 100);

  // Create a p5.MediaElement using createAudio().
  dragon = createAudio('/assets/lucky_dragons.mp3');

  // Show the default media controls.
  dragon.showControls();
```

Looking for the new version? Find p5.js 2.0 here!

Entry Header

Name: Nia Brathwaite

Role(s): Programmer

Primary responsibility for this work: Modify the code to implement features in line with the Side Quest requirements

Goal of Work Session

- Create Debug Screen
- Implement Moon Gravity feature
- Implement Speed Boost Feature

Tools, Resources, or Inputs Used

- GenAI - GitHub Copilot Chat (Version 0.38.2)
- Dr Karen Cochrane & David Han's Prior Code from GBDA 302 Week 9 Example 3 Folder
- P5.js reference library

GenAI Documentation

Date Used: March 23rd 2026

Tool Disclosure: GitHub Copilot Chat (Version 0.38.2)

Purpose of Use: Debugging

Summary of Interaction: Fixed the speed boost error in my code, as it was not working.

Human Decision Point(s): I did not overrule the suggestions made by GenAI as I needed help for the implementation, and my code was not working. Copilot showed the errors to fix the implementation.

Integrity & Verification Note: I tested the implementation of the fix to ensure it was in line with the Side Quest requirements and the feature I was aiming for. I also double-checked the P5.js reference library to ensure everything was correct.

Scope of GenAI Use: GenAI fixed two lines in correcting my code for the implementation of the speed boost and moon gravity; all other code was written by me or taken from the GBDA 302 Week 9 Example by Dr Karen Cochrane & David Han.

Limitations or Misfires: The tool could not account for the course concepts and my vision for the feature implementation.

Summary of Process (Human + Tool)

- Testing the Moon gravity feature
- Realised it needed constraints to keep the fox in the canvas
- Added constraints
- Tried to implement the Speed Bost Feature
- Did not work
- Asked GenAI for help
- Implemented GenAI suggestions.

Decision Points & Trade-offs

- I had considered adding barriers, but I felt that choice wasn't in line with the requirements.
- I chose speed as it is a common feature found in many games

Verification & Judgement

- Reread Side Quest Criteria
- Checked the p5.js reference library

Limitations, Dead Ends, or Open Questions

- The initial speed boost feature did not work properly, so I needed help from GenAI

Appendix

Prompt: I am trying to follow these instructions: add a debug screen to that allows you to toggle moon gravity on/off, and at least one other potential feature you might need in developing a game. I edited the skektch js to add a debug screen but my speed boost isnt working, how do i fix it and have i implement the moon gravity properly?

Output:

Optimizing tool selection...

Read [](file:///Users/niaalexis/Desktop/nbrathwa_sidequest_W9/sketch.js#1-1), lines 1 to 200

Read [](file:///Users/niaalexis/Desktop/nbrathwa_sidequest_W9/sketch.js#200-200), lines 200 to 263

I found the issues! Let me fix them:

Problem 1: Speed Boost Toggle — In `keyPressed()`, you have just `speedBoost;` which doesn't do anything. It needs to be `speedBoost = !speedBoost;`

Problem 2: Moon Gravity Value — You should use the `MOON_GRAVITY_VALUE` constant instead of hardcoding `2` for consistency.

Replacing 8 lines with 8 lines in [](file:///Users/niaalexis/Desktop/nbrathwa_sidequest_W9/sketch.js)

```

```

Fixes Applied:

1. **Speed Boost** — Changed `speedBoost;` to `speedBoost = !speedBoost;` in the keyPressed function. Now pressing "B" will correctly toggle the speed boost on/off.

2. **Moon Gravity** — Changed the hardcoded `2` to use the `MOON_GRAVITY_VALUE` constant for consistency and maintainability. Your moon gravity implementation is now correct—it toggles properly with the "G" key and adjusts jump velocity accordingly.

Your debug screen implementation looks good! The display shows both features' states correctly.

Made changes.